



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Sub'ect	DSA-2
Academic Year	1-III 2025-26 Semester
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Branch	CSIT
Course Outcome	CO-6

Case Study Topic: Content Search Optimization using Hashing

Work Done Summary:

In this case study, we implemented Hashing techniques for fast content retrieval in the Media Vault system. Digital content records such as movies, web series, and videos were stored using unique Content IDs as keys. Hash Tables enabled quick searching and retrieval of content information.

The system performed insertion, searching, updating, and deletion operations efficiently. Hashing significantly reduced search time compared to linear search methods, making content access faster for users.

Implementation Details:

- 1) Created a Hash Table using Content ID as the key.
- 2) Inserted digital content records into the Hash Table.
- 3) Implemented search operation for content retrieval.
- 4) Implemented update operation for content metadata.
- 5) Implemented deletion of unavailable content.
- 6) Handled collisions using chaining technique.
- 7) Compared hashing performance with linear search.
- 8) Verified near $O(1)$ average search complexity.

Result: Screenshots / Output

```
1  import java.util.HashMap;
2  import java.util.Scanner;
3  public class MediaVaultHashing {
4      public static void main(String[] args) {
5          HashMap<Integer, String> contentMap = new HashMap<>();
6          contentMap.put(101, "Avengers");
7          contentMap.put(102, "Stranger Things");
8          contentMap.put(103, "Money Heist");
9          contentMap.put(104, "Wednesday");
10         System.out.println("Available Content:");
11         for (Integer id : contentMap.keySet()) {
12             System.out.println(id + " -> " + contentMap.get(id));
13         }
14         Scanner sc = new Scanner(System.in);
15         System.out.print("\nEnter Content ID to Search: ");
16         int searchId = sc.nextInt();
17         if (contentMap.containsKey(searchId)) {
18             System.out.println("Content Found: " + contentMap.get(searchId));
19         } else {
20             System.out.println("Content Not Found");
21         }
22         sc.close();
23     }
24 }
```

Output:-

```
1  Available Content:
2  101 -> Avengers
3  102 -> Stranger Things
4  103 -> Money Heist
5  104 -> Wednesday
6
7  Enter Content ID to Search: 105
8  Content Not Found
```

The Hashing-based system provided fast and efficient content retrieval. Compared to traditional searching methods, Hash Tables significantly improved search speed and enhanced the user experience on the Media Vault platform..

GitHub Link: <https://github.com/snehiths399-byte/MediaVault--Digital-content-streaming-and-Analytics-system/commit/e13badaff9be0149df0d460482f0dd91139e5ab5>

Student Signature

Faculty Signature